VAAYU INTELLIGENCE PVT. LTD.

AIRSAFE

Know the air. Prove the data.

Real-time transmission-risk monitoring for hospitals, hotels and workplaces, with data anyone can verify.

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THE PROBLEM

Indoor air is the one risk nobody can see

Invisible

Exhaled air builds up when ventilation falls short. CO_2 reveals it, yet most buildings never measure it room by room.

Unmanaged

Teams react to complaints, not data. Ventilation follows a schedule, not how each room is actually used.

Unprovable

After an incident, logs can be edited and nobody outside the operator can check them.

THE SOLUTION

One system, from sensor to proof

A low-cost sensor node, a transparent risk engine and a verifiable data ledger in one platform.

1. Measures CO_2, particles, temperature, humidity, VOC and presence in every room
2. Scores transmission risk from 0 to 100 using published science
3. Explains why: score drivers, estimated air changes per hour, minutes above CO_2 and PM limits
4. Proves the data was never changed after it was recorded

HOW IT WORKS

From sensor to verified insight

1

Sense

ESP32 node: MH-Z19B CO_2, PMS5003 particles, SHT31 climate, SGP30 VOC, LD2410 radar presence

2

Send

Wi-Fi every 30 s, SD-card buffer through outages, each reading Ed25519-signed

3

Analyse

Rust (Axum) API on PostgreSQL: risk score, alerts, ventilation estimate

4

Prove

RFC 6962 Merkle ledger, signed checkpoints, root anchored on an EVM chain

THE SCIENCE

Transparent and cited, not a black box

Rudnick & Milton, 2003

Rebreathed-air fraction f = (CO_2 − 425) ÷ (38,000 − 425), taken from room CO_2

WHO guidelines, 2021

PM2.5 above 15 µg/m³ adds up to 15 points to the score

Every score explains itself

Drivers, values and sources are listed. Bands: Safe, Low, Moderate, High, Critical at 20, 40, 60, 80

0–100

Transmission Risk Index

CO_2 index × 0.85 plus PM2.5 and isolation-pressure penalties. An indicator, not a diagnosis.

ONE PLATFORM, THREE MARKETS

Built where air quality matters most

Hospital

AIRS Command Center: risk index per ward, isolation-room pressure and a live alert feed.

Hotel

Staff operations panel: room air-quality score, flagged tasks and occupancy.

Corporate

Workplace dashboard: space use, meeting no-shows and CO_2 reference status.

The live demo covers 62 rooms across all three markets.

WHAT SETS US APART

Data you can prove, not just trust

- Devices sign every reading with their own Ed25519 key (did:key identity)
- Readings are sealed in an RFC 6962 Merkle tree and each batch root is signed
- The root is anchored on a public chain: a timestamp no one can rewrite

Verify this reading

✓ Reading matches its ledger leaf

✓ Signed by the device key

✓ Inclusion proof reaches the batch root

✓ Checkpoint signed by the server key

Proves integrity and origin, not sensor calibration.

BUSINESS MODEL

Hardware paid once, software paid monthly

Sensor node

One-time hardware for each monitored room or zone.

Platform subscription

Per room per month: dashboards, alerts, analysis and reports.

Verified reports

Audit-ready proof for compliance and incident review.

Start with pilot wards and hotel floors, then expand to offices. Pricing is agreed per pilot.

WHERE WE ARE

Platform built, hardware in validation

BUILT AND TESTED

- Risk engine checked against 504 reference cases
- Ventilation estimate recovers known rates within 8% in simulation
- Crypto and ledger checked against published vectors (EIP-155, RFC 6962), 33 automated tests

IN PROGRESS

- Bench validation of the ESP32 node with real sensors
- Calibration of pressure and airflow sensors
- Pilot-site planning

NEXT

- Pilot on one ward or hotel floor
- Independent sensor validation
- Alerts to phones and building systems

TEAM AND ASK

The people, and what we need

Shubhika

Founder

Leads Vaayu Intelligence and the AIRSAFE vision.

Dinesh Kumar

Co-founder

Cybersecurity analyst (SOC). SIH 2024 and KAVACH 2023 finalist.

What we are asking for

A pilot site: ward, hotel floor or office

Clinical and facilities advisors to review thresholds

Funding and lab access for sensor calibration

Thank you. Questions welcome.